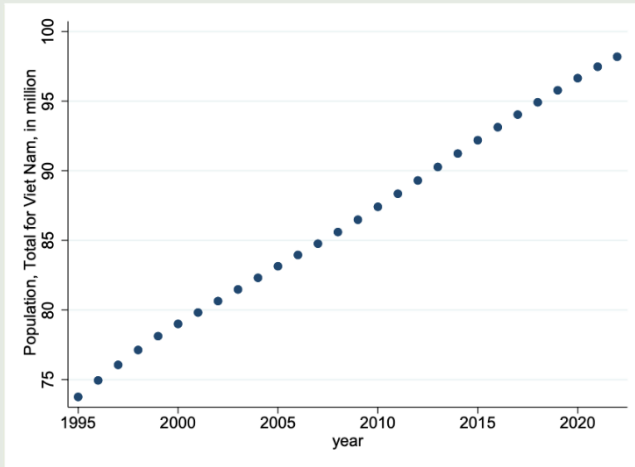


RANDOM PROCESSES

MAIN CONTENT

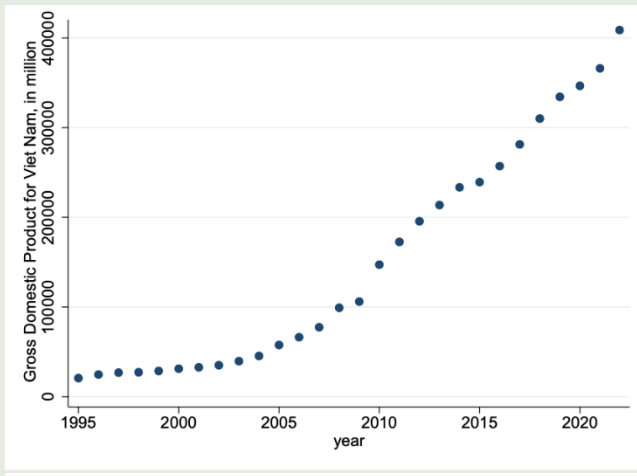
- White noise.
- Random walk process.
- Autoregressive process.
- Moving average process.
- Mean function.
- Measure of dependence.

Example



What functional form does the trend of this data follow?

Example



What functional form does the trend of this data follow?

It is necessary to explore the properties of random processes in order to solve some related problems. For example:

Illustrative example

If we need to find an estimating model with certain properties to solve the problem of autocorrelation, which process is appropriate?

- To solve such a problem, it is necessary to relate the random processes of quantitative phenomena to the observations we have.
- The basic random processes in this lesson are the way to help us solve problems with current observations based on past observations or past random phenomena.
- This approach is called the time domain approach.

White noise

Random walk process

Autoregressive processes

Moving Average (MA) Processes

Expected function

Measuring Dependency

White noise

- A sequence of random variables, such as $x_1, \dots, x_n$ (or possibly an infinite sequence $\dots, x_{-1}, x_0, x_1, \dots$) is often said to be dependent of each other.
- A process consisting of a sequence of independent random variables, having zero mean, constant variance, and indexed over time, is often called **white noise**.

White noise

A sequence of random variables $x_1, x_2, \dots, x_n$ is called **white noise** if

$$E(x_t) = 0$$

$$\text{Var}(x_t) = \sigma^2 \text{ (variance is a constant)}$$

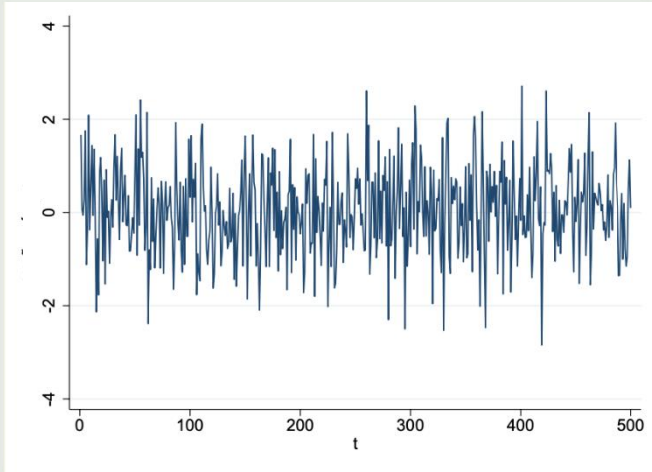
$$\text{Cov}(x_s, x_t) = 0 \text{ for all } s \neq t$$

- A particular example is **Gaussian white noise**; denoted as $w_1, w_2, \dots, w_n$.
- These random variables are called **Gaussian white noise**, if all w_t are random variables that follow a normal distribution and are independent of each other, that is, $w_t \sim N(0, \sigma_w^2)$.

Note: Uncorrelated random variables do not mean they are independent. However, if random variables are independent, they will be uncorrelated with each other.

Simulation of Gaussian white noise process

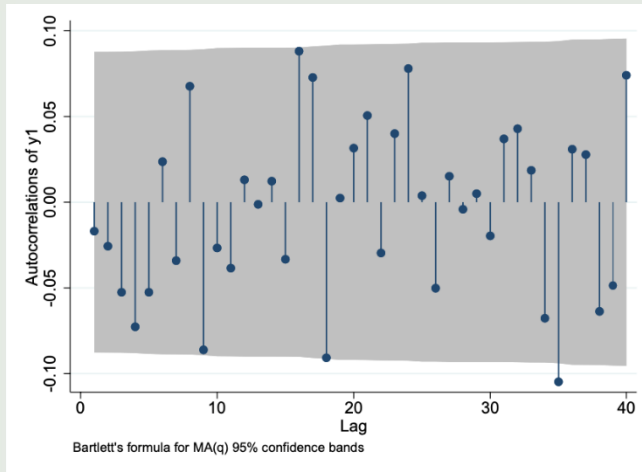
Example



Any comments on this process?

Simulation of Gaussian white noise process

Example



Any comments on this graph?

Simulation of Gaussian white noise process

```
. corrgram y1, lags(16)
```

LAG	AC	PAC	Q	Prob>Q	-1 [Autocorrelation]	0 [Partial autocor]	1 -1 0 1
1	-0.0169	-0.0169	.14345	0.7049			
2	-0.0256	-0.0260	.475	0.7886			
3	-0.0525	-0.0537	1.8694	0.6000			
4	-0.0727	-0.0759	4.5429	0.3375			
5	-0.0525	-0.0596	5.9426	0.3118			
6	0.0236	0.0145	6.2265	0.3983			
7	-0.0341	-0.0450	6.8177	0.4481			
8	0.0677	0.0556	9.1535	0.3295			
9	-0.0861	-0.0942	12.945	0.1651			
10	-0.0267	-0.0324	13.31	0.2068			
11	-0.0384	-0.0446	14.068	0.2292			
12	0.0131	0.0023	14.156	0.2908			
13	-0.0012	-0.0147	14.157	0.3629			
14	0.0123	-0.0126	14.235	0.4324			
15	-0.0333	-0.0374	14.808	0.4653			
16	0.0881	0.0758	18.834	0.2774			

Any comments on this graph?

White noise

Random walk process

Autoregressive processes

Moving Average (MA) Processes

Expected function

Measuring Dependency

Random walk

Example

A simple way to model forward movement is to use the equation as follows:

$$x_t = x_{t-1} + 1$$

with $t = 1, \dots, n$.

What does this model tell?

Random walk

- One process commonly used for trend analysis is **the random walk process**.
- It can be understood as "someone's current position is determined by their position at the last step plus the random step they just took".
- So the equation of the random walk has the form:

$$x_t = x_{t-1} + w_t \quad (1)$$

with $t = 1, \dots, n$ and w_t are Gaussian white noises.

Random walk

- A (non-random) sliding term, δ , can be added to the above process, then:

$$x_t = \delta + x_{t-1} + w_t \quad (2)$$

- Equations (1) and (2) can be expressed as:

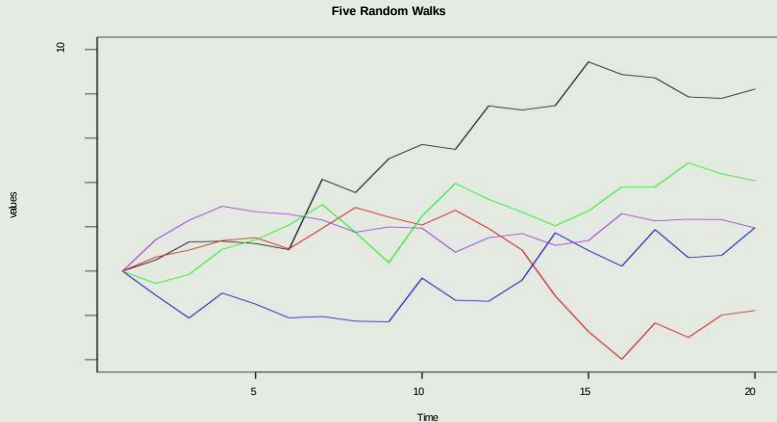
$$x_t = \sum_{i=1}^t w_t \quad (3)$$

- For the case with sliding terms:

$$x_t = \delta t + \sum_{i=1}^t w_t \quad (4)$$

Random walk

Example



Comments?

Random walk

- If a random walk consists of a sequence of dependent random variables, it can be transformed into an independent sequence by taking “**differences**”.
- Depending on the characteristics of the time series, it is possible to take the difference once or many times to get an independent series.
- The difference taken once is called the first **order difference** , calculated by the formula:

$$\nabla x_t = x_t - x_{t-1}, \quad (5)$$

with $t = 1, \dots, n - 1$.

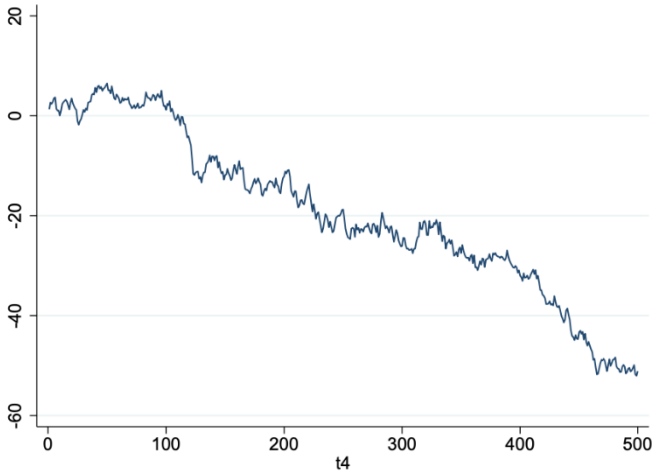
- And **the second difference** is calculated by the formula:

$$\nabla^2 x_t = \nabla x_t - \nabla x_{t-1}, \quad (6)$$

with $t = 1, \dots, n - 1$.

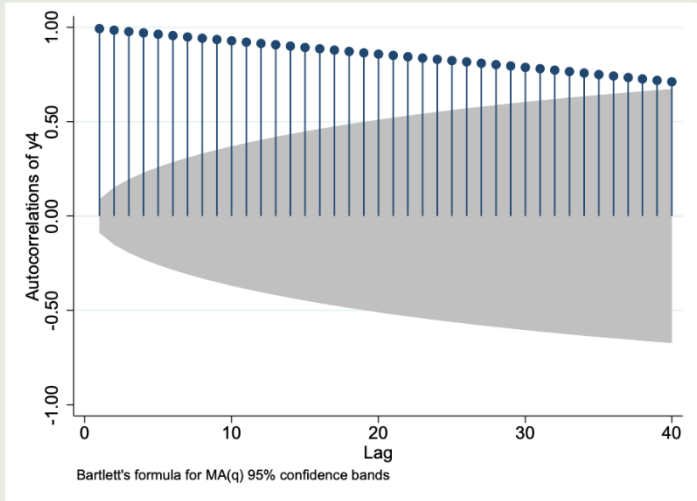
Random walk simulation

Example



Random walk simulation

Example



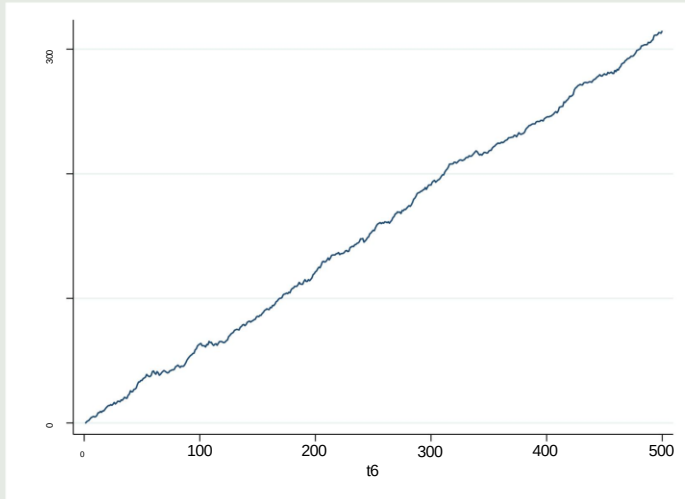
Random walk simulation

```
. corrgram y4, lags(16)
```

LAG	AC	PAC	Q	Prob>Q	-1 0 1 [Autocorrelation]	-1 0 1 [Partial autocor]
1	0.9926	1.0003	495.56	0.0000		
2	0.9847	0.0145	984.28	0.0000		
3	0.9773	0.1026	1466.7	0.0000		
4	0.9703	0.0045	1943.1	0.0000		
5	0.9630	0.0445	2413.3	0.0000		
6	0.9556	-0.0030	2877.3	0.0000		
7	0.9486	0.0386	3335.4	0.0000		
8	0.9418	0.0330	3787.9	0.0000		
9	0.9349	-0.0198	4234.7	0.0000		
10	0.9283	0.1164	4676.1	0.0000		
11	0.9211	-0.0749	5111.6	0.0000		
12	0.9141	0.0483	5541.4	0.0000		
13	0.9073	0.0414	5965.8	0.0000		
14	0.9003	0.0075	6384.4	0.0000		
15	0.8931	0.0257	6797.2	0.0000		
16	0.8859	0.0478	7204.2	0.0000		

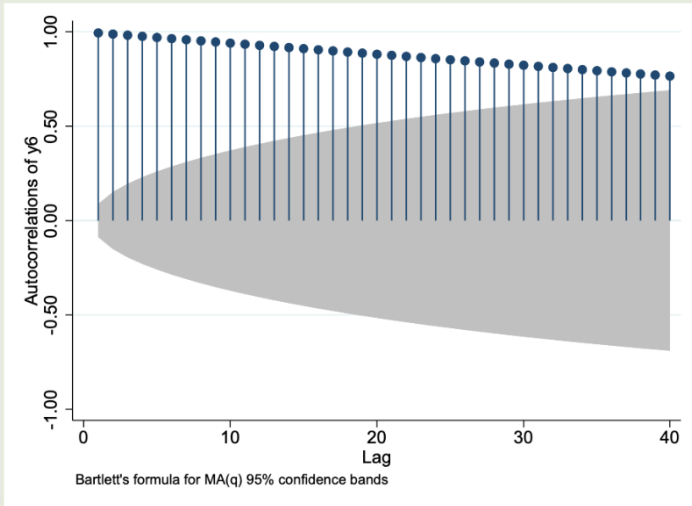
Random walk simulation with sliding factor of 0.7

Example



Random walk simulation with sliding factor of 0.7

Illustrative example



Random walk simulation with sliding factor of 0.7

```
. corrgram y6, lags(16)
```

LAG	AC	PAC	Q	Prob>Q	-1	0	1	-1	0	1
					[Autocorrelation]			[Partial autocor]		
1	0.9940	1.0001	496.96	0.0000						
2	0.9880	0.0362	988.96	0.0000						
3	0.9820	0.0519	1476	0.0000						
4	0.9760	0.0389	1958.1	0.0000						
5	0.9700	-0.0758	2435.2	0.0000						
6	0.9641	-0.0032	2907.4	0.0000						
7	0.9581	0.0307	3374.8	0.0000						
8	0.9521	0.0012	3837.2	0.0000						
9	0.9461	-0.0675	4294.9	0.0000						
10	0.9402	-0.0839	4747.7	0.0000						
11	0.9343	0.0469	5195.7	0.0000						
12	0.9284	0.0796	5639	0.0000						
13	0.9225	0.0083	6077.7	0.0000						
14	0.9167	0.0545	6511.6	0.0000						
15	0.9107	0.0106	6940.9	0.0000						
16	0.9048	0.0267	7365.5	0.0000						

White noise

Random walk process

Autoregressive processes

Moving Average (MA) Processes

Expected function

Measuring Dependency

Autoregressive processes

A class of processes closely related to random walks is **the autoregressive (AR) process**.

An autoregressive process is defined in the sense that "the current position is a linear combination of previous positions plus a random term (in the form of Gaussian white noise)".

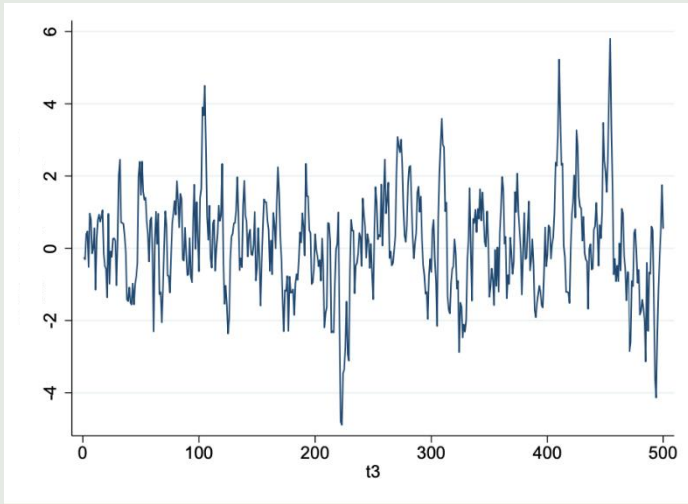
$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \cdots + \phi_p x_{t-p} + w_t \quad (7)$$

This process is called **an AR(p) process**.

Under what conditions can a random walk be considered a special case of an AR process?

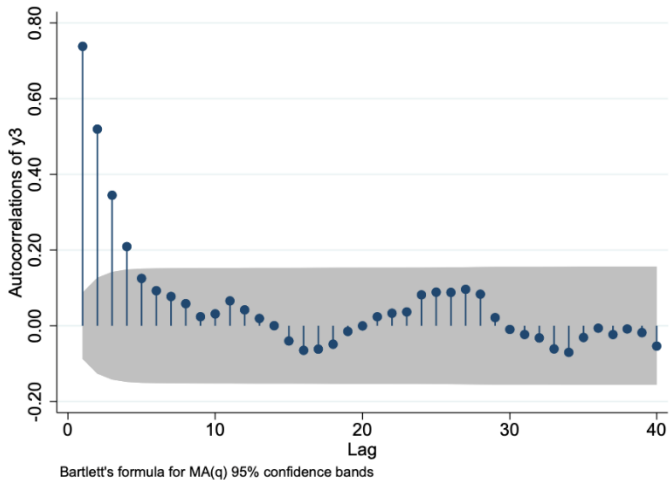
Simulation of AR(1) process with coefficient equal to 0.8

Example



Simulation of AR(1) process with coefficient equal to 0.8

Example



Simulation of AR(1) process with coefficient equal to 0.8

```
. corrgram y3, lags(16)
```

LAG	AC	PAC	Q	Prob>Q	-1	0	1	-1	0	1
					Prob>Q	[Autocorrelation]	[Partial autocor]			
1	0.7381	0.7382	274.02	0.0000						
2	0.5195	-0.0548	410.02	0.0000						
3	0.3447	-0.0429	470.04	0.0000						
4	0.2092	-0.0355	492.19	0.0000						
5	0.1250	0.0099	500.11	0.0000						
6	0.0925	0.0440	504.46	0.0000						
7	0.0772	0.0110	507.5	0.0000						
8	0.0583	-0.0172	509.23	0.0000						
9	0.0238	-0.0452	509.52	0.0000						
10	0.0314	0.0707	510.02	0.0000						
11	0.0656	0.0753	512.24	0.0000						
12	0.0419	-0.0966	513.14	0.0000						
13	0.0191	-0.0183	513.33	0.0000						
14	0.0001	-0.0075	513.33	0.0000						
15	-0.0402	-0.0550	514.17	0.0000						
16	-0.0649	-0.0072	516.35	0.0000						

White noise

Random walk process

Autoregressive processes

Moving Average (MA) Processes

Expected function

Measuring Dependency

Moving Average Processes (MA)

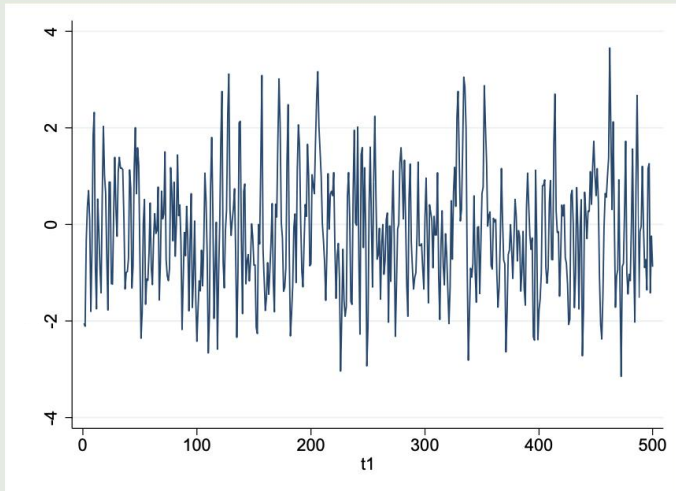
Consider the white noise process, $w_1, \dots, w_n, \dots$, then a **moving average process is represented as:**

$$x_t = w_t + \theta_1 w_{t-1} + \dots + \theta_q w_{t-q} \quad (8)$$

This process is called **the MA(q) process**.

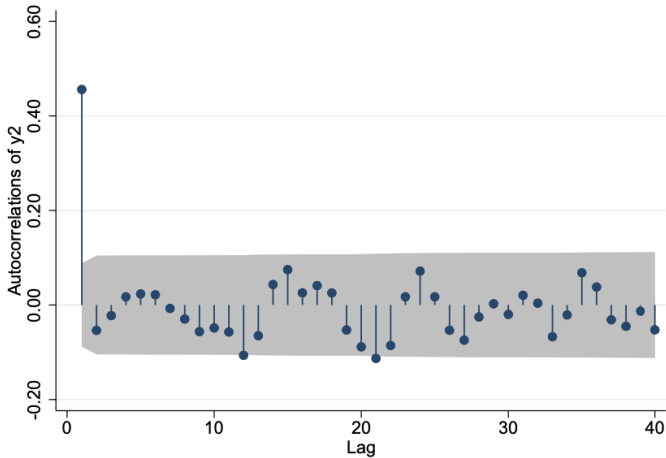
Simulation of MA(1) process with coefficient equal to 0.8

Example



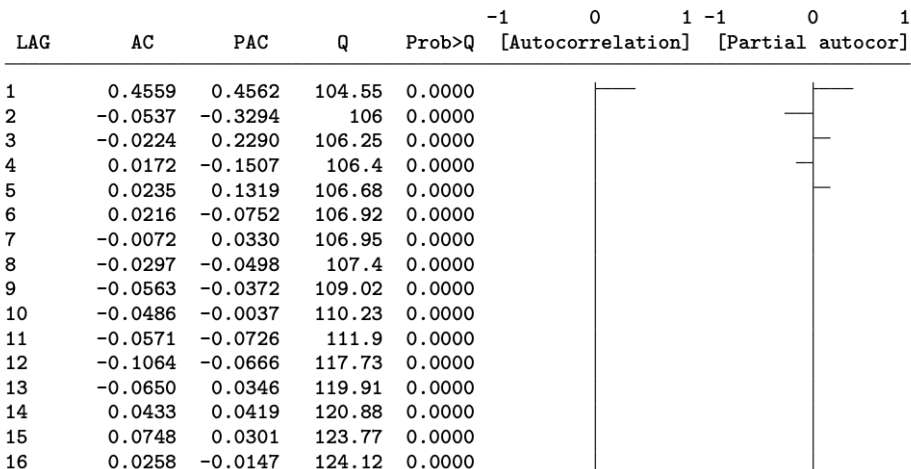
Simulation of MA(1) process with coefficient equal to 0.8

Example



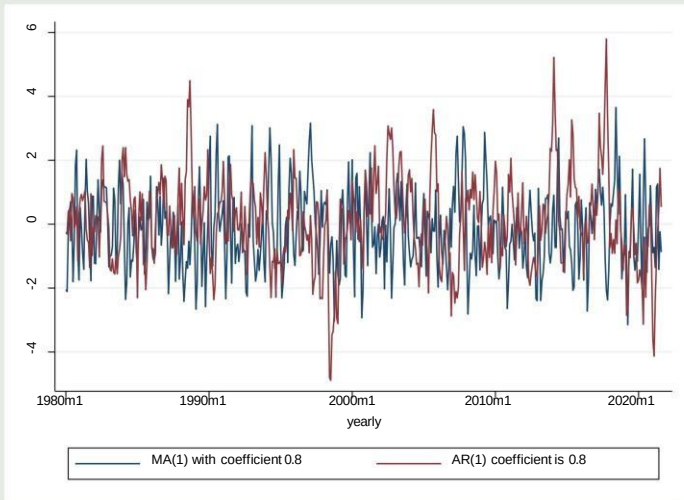
Simulation of MA(1) process with coefficient equal to 0.8

```
. corrgram y2, lags(16)
```



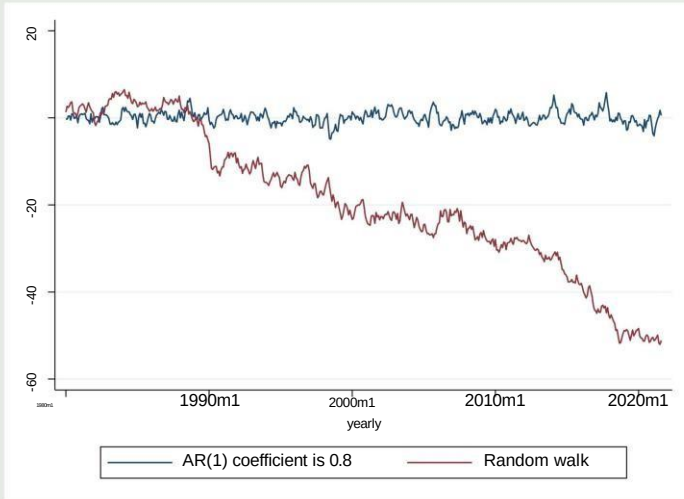
Simultaneous graph of series

Example



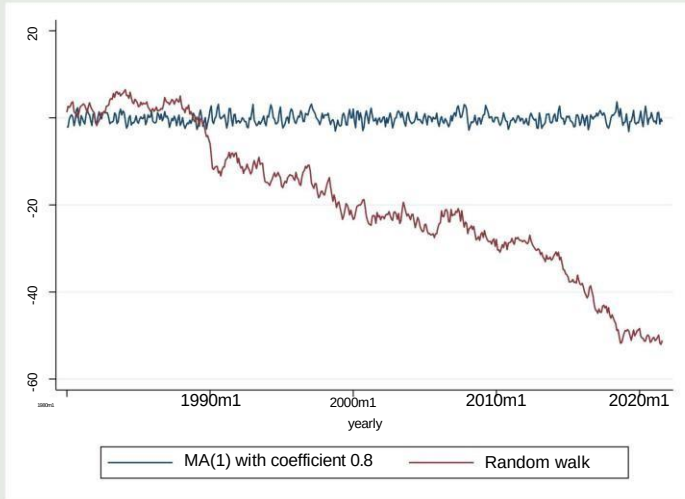
Simultaneous graph of series

Example



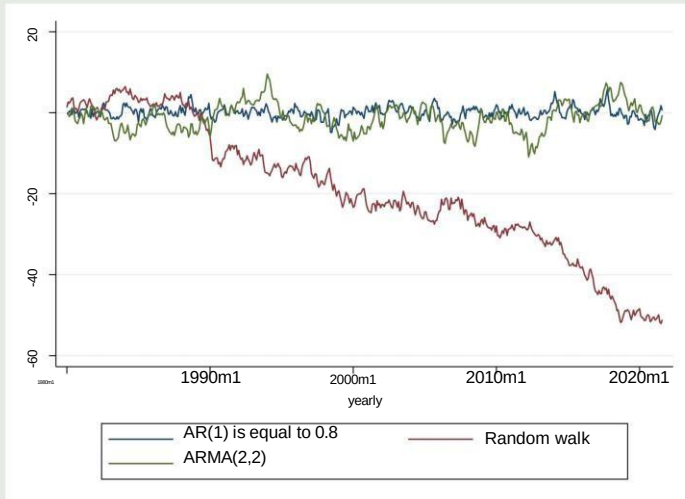
Simultaneous graph of series

Example



Simultaneous graph of series

Example



White noise

Random walk process

Autoregressive processes

Moving Average (MA) Processes

Expected function

Measuring Dependency

The expectation function (**mean function**) is defined as:

$$\mu_t = E(x_t) \quad (9)$$

Example

Calculate the expectation of:

- Random walk with sliding term.
- AR(1) autoregressive process
- MA(1) Process

White noise

Random walk process

Autoregressive processes

Moving Average (MA) Processes

Expected function

Measuring Dependency

Variance and covariance

The variance of a random variable X is calculated by the formula:

$$\text{Var}(X) = E[(X - EX)^2] = E(X^2) - (EX)^2 \quad (10)$$

The covariance of two random variables, X and Y is calculated by the formula:

$$\text{Cov}(X, Y) = E[(X - EX)(Y - EY)] \quad (11)$$

The correlation coefficient of two random variables, X and Y , is calculated by the formula:

$$\text{Corr}(X, Y) = \rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \quad (12)$$

Uncorrelated random variables

- Two random variables X and Y are considered uncorrelated when $\rho(X, Y) = 0$.
- Two random variables X and Y are independent when their density functions are both equal to the product of their two marginal density functions.
- If X and Y are independent, then they are also uncorrelated.
- However, if X and Y are uncorrelated, they can still be dependent on each other.

Autocovariance function

A common feature of time series is that observations are dependent on each other. The **autocovariance** function is defined as follows:

$$\gamma(t, s) = E[(x_s - \mu_s)(x_t - \mu_t)] \quad (13)$$

- Thus, this is simply the covariance between x_s and x_t considered with all s and t .
- Covariance measures the strength of linear dependence between random variables.
- A small covariance for s, t close together often shows random variables that closely resemble white noise.
- Smooth series are likely to have large covariance even for with s and t far apart.

Example

Calculate the autocorrelation of:

- Random walk.
- MA(2) process
- AR(2) process

Autocorrelation function

The autocorrelation function (**ACF**) is defined as:

$$\rho(s, t) = \frac{\gamma(s, t)}{\sqrt{\gamma(s, s)\gamma(t, t)}} \quad (14)$$

Illustrative example

From the simulation results of the processes, what is the ACF function used for?